Composition Pair B1.13 + B1.01: The Enabling Pair Where Instinct/Reasoning Separation Makes Mutation Architecturally Meaningful by Defining What LLM Updates Affect and Preserve, and Mutation Governance Instruments Are Needed Precisely Because the Instinct Layer Can Change Independently of the Reasoning Layer

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Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026), the second paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026). It does not introduce new axioms. Its contribution is to formalize, in composition-pair form, the mutual enabling relationship between B1.13 (mutation as instinct-layer evolution) and B1.01 (instinct/reasoning separation), establishing that the separation is what makes mutation architecturally specific and that the mutation governance instruments are motivated precisely by the independence the separation creates.

Abstract

B1.13 and B1.01 stand in an enabling pair relationship: each depends on the other for its architectural significance, and neither is fully meaningful without the other present. B1.01 (the instinct/reasoning separation) makes B1.13 (mutation) architecturally specific by defining what an LLM vendor update affects—the instinct layer—and what it preserves—the reasoning layer (DNA). Without B1.01's separation, LLM updates are undifferentiated system changes; with B1.01 in place, an LLM update is a specific instinct-layer event that leaves the reasoning layer independent, creating a misalignment risk that B1.13's three governance instruments are designed to address. The enabling direction also runs in reverse: B1.01 motivates every design choice in B1.13's instrument set. Verification gates are meaningful only when there is a separate reasoning layer to verify against. Routing—controlling which cells use which LLM version—is meaningful only when different instinct-layer versions can coexist alongside a stable reasoning layer. Pinning high-stakes decisions to reasoning-layer rules (DNA) is only possible when the reasoning layer is architecturally independent of the instinct layer. Beyond the enabling direction, the pair exhibits a preservation function: B1.01 establishes the instinct/reasoning separation at deployment, and B1.13 maintains it through evolution. Without B1.13, the separation is correct at deployment time but degrades through ungoverned LLM updates. Together, B1.01 and B1.13

describe a dynamic separation—not a one-time architectural property but a continuously governed relationship. Failure modes are symmetric: B3.14 (Ungoverned Mutation) when B1.13 is absent while B1.01 is present, and B3.02 (Instinct-Reasoning Collapse) when B1.01 is absent or eroded.

1. Pair Identification

Pair: B1.13 (mutation—instinct-layer evolution through LLM vendor updates, governed through verification gates, routing, and pinning) and B1.01 (the instinct/reasoning separation—the architectural commitment that the instinct layer, occupied by the LLM, and the reasoning layer, occupied by the DNA substrate, are independently evolvable and independently governed).

Relationship type: Enabling pair with a preservation function. The enabling direction is primarily B1.01 → B1.13: B1.01 makes B1.13 architecturally specific by defining the target of mutation (instinct layer) and the thing mutation preserves (reasoning layer). A secondary enabling direction runs in the opposite orientation: B1.01 motivates each of B1.13's three governance instruments by providing the architectural asymmetry those instruments are designed to manage. The preservation function is the pair's temporal dimension: B1.01 establishes the separation; B1.13 maintains it through the LLM updates that would otherwise erode it.

Source sections: Paper 2 §4 (instinct/reasoning separation, B1.01), §7.2 (instinct evolution as mutation, B1.13), §8.2 (per-mechanism governance shape), §8.3 (the instinct/reasoning boundary as governed substrate content).

Position in the note series: This is B4.18, the eighteenth Phase B4 composition pair, the seventeenth composition pair following B4.01. The preceding B4 notes (B4.01–B4.17) are complete. The succeeding note, B4.19, covers the B1.06 + B1.14 enabling pair (two-layer structure and directed selection).

2. The Enabling Direction: B1.01 Makes Mutation Architecturally Specific

Without the instinct/reasoning separation, an LLM vendor update is an undifferentiated change to the system. No architectural distinction identifies which layer changed, which layer stayed the same, or what the implications of the change are. The system has changed; whether that change is good or bad, large or small, compatible or incompatible with prior behavior—these are empirical questions with no architectural frame.

B1.01 changes this. Once the instinct/reasoning separation is in place, the LLM vendor update is no longer undifferentiated. It is a specific instinct-layer event. The instinct layer—the LLM, the fast-pattern, inference-at-runtime component—has changed. The reasoning layer—the DNA, the stabilized orchestration substrate, the governed human-authored rules—has not changed. B1.01 is what permits this sentence to be stated: LLM updates affect the instinct layer; the reasoning layer remains independent.

This specificity is the enabling condition for mutation as an architectural concept. Mutation in B1.13 is not merely the colloquial fact that LLMs get updated. Mutation is the event in which the instinct layer changes while the reasoning layer does not—and the architectural significance of that event is entirely a product of B1.01's separation. A system that has not established the instinct/reasoning separation cannot have mutation in the B1.13 sense; it can have LLM updates, but those updates do not distinguish instinct from reasoning, and no architectural governance is required in response to the distinction that does not exist.

B1.01 also names what mutation creates: misalignment risk. When the LLM updates and the instinct layer changes, the DNA's rules and specifications—which were written with reference to prior instinct-layer behavior—may no longer accurately describe or constrain what the new instinct layer does. The instinct layer's outputs may no longer align with what the reasoning layer specifies those outputs should be. This is not a failure of deployment; it is an architectural feature of having independently evolvable layers. The misalignment risk is the price of independent evolvability, and it is precisely the thing B1.13's governance instruments exist to manage.

None of this is articulable without B1.01. The concept of instinct-reasoning misalignment requires an instinct/reasoning distinction. The concept of instinct-layer verification requires a reasoning-layer standard to verify against. The concept of preserving the reasoning layer through an LLM update requires that the reasoning layer be architecturally distinct from the LLM in the first place. B1.01 is not a prerequisite for B1.13 in a merely logical sense; B1.01 is the source of B1.13's architectural meaning.

3. The Motivating Direction: B1.01 Motivates Each of B1.13's Three Governance Instruments

B1.13's mutation governance instruments—verification gates, routing, and pinning—are not generic change-management tools. Each is specifically designed for a separated architecture, and each is motivated by a different aspect of the instinct/reasoning separation B1.01 establishes.

Verification gates are the instrument through which the integration of an LLM update is assessed against the reasoning layer. When a new LLM version arrives, verification gates test whether the new instinct-layer behavior still aligns with the reasoning-layer specifications—whether the new LLM does what the DNA rules say a cell governed by those rules should do. The verification is instinct-against-reasoning: the governance instrument compares the new instinct layer's output to the independent standard the reasoning layer supplies.

This instrument is only meaningful when a separate reasoning layer exists. Without B1.01's separation, there is no independent standard to verify against. A deployment without the instinct/reasoning separation might test a new LLM version against prior LLM outputs—but that is a behavioral regression test, not architectural verification. Architectural verification requires an independent standard; the reasoning layer is that standard; B1.01 is what makes the reasoning layer independent. Verification gates in B1.13's sense presuppose B1.01's sense.

Routing is the instrument through which cells are assigned to specific LLM versions following a mutation event. Not every cell needs to adopt a new instinct layer simultaneously; routing determines which cells use the new version, which continue on the prior version, and—in parallel-run patterns—which run both for comparative verification. Routing is meaningful because different LLM versions (instinct layers) can coexist alongside a single stable reasoning layer. The reasoning layer's rules govern the cells regardless of which instinct-layer version they use; the reasoning layer does not change when the LLM changes. This stability is what makes parallel operation under routing possible.

Without B1.01's separation, changing the LLM changes the cell uniformly. There is no reasoning layer that remains stable across instinct-layer versions to hold the cell's governance architecture together. Routing as an instrument—assigning different cells to different LLM versions while maintaining coherent governance—requires a stable independent reasoning layer to which all routed cells equally answer. That stable independent reasoning layer is what B1.01 establishes.

Pinning is the instrument through which high-stakes decisions are architecturally anchored to reasoning-layer rules regardless of how the instinct layer changes. A cell handling high-stakes decisions is configured so that those decisions remain governed by explicit DNA rules rather than by instinct-layer inference, even after an LLM update. The instinct layer may become more capable; pinned decisions remain under reasoning-layer governance until humans explicitly relocate the boundary.

Pinning is only possible when the reasoning layer is architecturally independent of the instinct layer. Pinning means: this decision goes to reasoning, not to instinct—and that sentence requires the two to be distinct. Without B1.01's separation, there is no independent reasoning layer to pin to. The concept of anchoring a decision to the reasoning layer while the instinct layer changes presupposes that the reasoning layer and the instinct layer are separate things, independently evolvable, with separate governance. B1.01 establishes that separation; pinning exploits it.

The three instruments together are the operational response to the misalignment risk B1.01 creates by establishing independent evolvability. Verification gates detect misalignment before it propagates. Routing controls the rate and scope at which the new instinct layer is adopted. Pinning protects the decisions most sensitive to misalignment from ever being governed by unverified instinct. All three instruments are architecturally coherent only because B1.01 has separated what they distinguish, stabilized what they rely on, and named what they protect.

4. Mutual Dependency

The enabling direction in §§2–3 runs from B1.01 to B1.13: B1.01 provides the architectural separation that makes mutation specific and provides the separated structure that makes the governance instruments meaningful. The dependency in the other direction is equally real.

B1.01 depends on B1.13 for the maintenance of its separation through evolution. The instinct/reasoning separation that B1.01 establishes is a property of the deployed architecture at a moment in time. LLM vendor updates will occur regardless of whether B1.13 is in place. When an LLM updates without B1.13's governance instruments, the instinct layer changes ungoverned: the alignment between the new instinct layer and the existing reasoning layer is not tested (no verification gates), the adoption scope is not controlled (no routing), and high-stakes decisions are not protected (no pinning). The separation between instinct and reasoning is formally preserved—the DNA still governs, the LLM is still the instinct layer—but the instinct layer's alignment with the reasoning layer's specifications drifts ungoverned. The anti-pattern this describes is B3.14 (Ungoverned Mutation): B1.01 is present, B1.13 is absent, and the separation degrades through each unverified LLM update.

This dependency names a vulnerability. B1.01 correctly identifies that the instinct and reasoning layers are independently evolvable; B1.13 is the governance response to what independent evolvability implies in practice. A deployment that establishes B1.01's separation at deployment time but does not implement B1.13's mutation governance has correctly described its architecture and incorrectly governed its evolution. Over time, through accumulated ungoverned LLM updates, the instinct layer may behave in ways the reasoning layer's rules do not anticipate, and the practical alignment between layers erodes even if the formal architectural distinction remains.

B1.13 depends on B1.01 for the instrument design. The three mutation governance instruments are designed for a separated architecture. Verification gates test instinct-layer behavior against reasoning-layer specifications; if there is no architectural separation, there is no independent specification to test against and the instrument has no function. Routing manages cells across LLM versions within a stable reasoning-layer governance structure; if there is no separation, there is no stable reasoning layer to hold across versions and the instrument has no anchor. Pinning redirects decisions from the instinct layer to the reasoning layer; if there is no separation, there is no distinct reasoning layer to redirect to and the instrument has no target. Without B1.01, the instruments would need to be redesigned from scratch or dropped as inapplicable.

The mutual dependency is not symmetrical in strength: the B1.01 → B1.13 direction (§§2–3) is the enabling direction, where B1.01 defines B1.13's architectural meaning. The B1.13 → B1.01 direction is the maintenance direction, where B1.13 preserves what B1.01 established. The pair requires both directions to be operationally complete.

5. Co-Presence Requirement and the Preservation Function

The mutual dependency in §4 implies a co-presence requirement: both B1.01 and B1.13 must be present for the architecture to maintain instinct-reasoning alignment through evolution. Examining each half-presence state clarifies why.

B1.01 without B1.13—instinct/reasoning separation without mutation governance. The separation is correctly established at deployment. The DNA layer governs the reasoning; the LLM occupies the instinct layer; the two are architecturally distinct. LLM vendor updates arrive. The instinct layer changes. No verification gates assess the alignment of the new instinct-layer behavior against the reasoning layer’s specifications. No routing manages the scope and pace of adoption. No pinning protects high-stakes decisions from the effects of unverified instinct-layer change. The separation remains formally present—the DNA still governs, the LLM is still the instinct layer—but the instinct layer’s alignment with the DNA’s specifications is no longer maintained through the update cycle. The system is in the B3.14 (Ungoverned Mutation) state: the architectural separation exists in name; the governed relationship the separation was supposed to sustain has eroded in practice.

B1.13 without B1.01—mutation governance without instinct/reasoning separation. Mutation governance instruments exist in the deployment. Verification gates are present; routing is present; pinning is present. But the architectural separation that gives these instruments their targets is absent. Verification gates: verifying instinct-layer behavior against what standard? Without a separate reasoning layer, there is no independent specification to verify against. Routing: assigning cells across LLM versions within what stable governance structure? Without a reasoning layer that persists across instinct-layer versions, there is no stable anchor. Pinning: redirecting high-stakes decisions to what? Without a separate reasoning layer, there is no independent destination. The instruments are present but architecturally untethered. This is the B3.02 (Instinct-Reasoning Collapse) orientation of the failure: mutation governance exists but the separation it was designed to govern does not.

Full co-presence. B1.01 establishes the instinct/reasoning separation: the DNA layer is the reasoning layer; the LLM is the instinct layer; the two are independently evolvable and independently governed. B1.13 governs how the instinct layer evolves within that separation: verification gates test each LLM update’s alignment with the reasoning layer; routing controls adoption scope; pinning protects high-stakes decisions. Together, the pair describes a **dynamic separation**—not a property established once and assumed to persist, but a governed ongoing architectural relationship maintained through each LLM update cycle.

This is the preservation function: B1.01 establishes the separation at deployment; B1.13 preserves it through evolution. The dynamic separation is the pair’s joint contribution to the architecture. Neither note delivers it alone. B1.01 alone delivers a separation that degrades ungoverned. B1.13 alone delivers instruments without a separated architecture to govern. Together they deliver a separation that is not merely stated but continuously maintained.

6. Operational Integration and Detection

Operational sequence. An LLM vendor update arrives—a mutation event in B1.13’s sense. The verification gates specified in B2.06 are invoked: the new instinct-layer behavior is evaluated against the existing reasoning-layer DNA specifications that B1.01 establishes as the independent standard. The evaluation may run in parallel-run pattern (new and prior LLM versions operating against the same cells), in retirement pattern (prior reasoning substrate retired where new instinct reliably handles its function), or in fallback-retention pattern (prior reasoning substrate retained as active verification on sensitive paths). Routing per B2.04 determines which cells adopt the new instinct layer based on the verification outcomes. High-stakes decisions under pinning per B2.05 remain governed by reasoning-layer DNA rules regardless of which instinct-layer version is routed to those cells—the pinning is to the reasoning layer as an architectural constant, not to a specific LLM version. Mutation governance records document how the instinct-reasoning separation was maintained through the update cycle, providing path retraceability for accountability over evolution events.

Detection criteria. A deployment can be assessed for full co-presence of B1.13 and B1.01 through three complementary tests.

> *B2.66 mutation governance verification* asks: are all three governance instruments—verification gates, routing, and pinning—in place and active for instinct-layer changes? If any instrument is absent, the deployment cannot demonstrate that LLM updates are being governed; the relevant anti-pattern is B3.14 (Ungoverned Mutation) or a partial governance failure depending on which instruments are missing.
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> *B2.03 architectural test* asks: does the instinct/reasoning separation hold after LLM updates? This is the test of B1.01's persistence through evolution rather than merely at deployment. Holding after updates requires B1.13's instruments to have been active during the update; the architectural test is the downstream evidence of the governance instruments having functioned.
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> *Pair completeness test* asks: do the mutation governance instruments specifically reference the instinct/reasoning separation in their design? Verification gates that verify instinct behavior against an independent DNA standard reference B1.01. Pinning instruments that anchor high-stakes decisions to DNA rules reference B1.01. If the governance instruments could be described without reference to the instinct/reasoning separation—if they are generic behavioral regression tests rather than separation-maintaining governance—the pair is not fully co-present even if both B1.01 and B1.13 are nominally present.

7. Failure Modes

Two failure modes correspond to the two half-presence states identified in §5. Both are named anti-patterns in the CKS derivation series, and both are reachable from a deployment that initially had the pair fully in place.

B3.14—Ungoverned Mutation. This failure mode arises when B1.01 is present and B1.13 is absent. The instinct/reasoning separation is correctly established. LLM updates occur. The instinct layer changes without governance: no verification against the reasoning layer, no controlled routing, no protected high-stakes decisions. The separation degrades not in its formal architectural statement but in its operational content: the reasoning layer continues to govern, but what it governs is a drifted instinct layer whose behavior the reasoning layer's specifications no longer accurately describe.

Ungoverned Mutation is reachable through attrition. A deployment that initially implemented B1.13 may allow governance instruments to fall into disuse—verification gates skipped because prior updates were unproblematic, routing collapsed to uniform adoption because the operational complexity seemed unnecessary, pinning removed because high-stakes paths were believed to have been thoroughly verified. Each individual decision to reduce governance overhead is defensible in isolation; the cumulative effect is Ungoverned Mutation.

B3.02—Instinct-Reasoning Collapse. This failure mode arises when B1.01 is absent or eroded. The architectural distinction between instinct and reasoning no longer holds: reasoning capacity has migrated into the LLM (into model weights rather than governance-accessible substrate), the LLM has been granted authority over substrate content it was designed to mediate, or the DNA layer has been allowed to become dependent on the LLM's outputs rather than independent of them. When B1.01 collapses, B1.13's mutation instruments become meaningless: there is nothing to verify the instinct layer against because there is no independent reasoning layer; there is no stable reasoning layer to route across LLM versions within; there is no separate reasoning layer to pin high-stakes decisions to.

Instinct-Reasoning Collapse is also reachable through incremental drift. A deployment may begin with B1.01 correctly established and allow the reasoning layer to erode—migration of orchestration logic into model context that does not survive LLM updates, informal patterns in which LLM outputs directly populate governance fields without passing through human-authored rules, gradual narrowing of what the DNA layer governs as the instinct layer proves reliable. The formal claim that the architecture has an instinct/reasoning separation persists; the operational content of that separation degrades.

The directional relationship between the two failure modes. B3.14 and B3.02 are not independent. Prolonged B3.14 (Ungoverned Mutation) can produce B3.02 (Instinct-Reasoning Collapse): if LLM updates proceed ungoverned long enough, the instinct layer's behavior may diverge from the reasoning layer's specifications to the point where the reasoning layer's rules become descriptively inaccurate, and the practical governance boundary between instinct and reasoning collapses even if the formal architectural distinction is nominally preserved. The governance instruments B1.13 provides are not just a response to the misalignment risk B1.01 creates; they are what prevents that misalignment risk from compounding into the architectural collapse B3.02 describes.

Source paper

Li, W. (2026). The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance. April 2026. ORCID: 0009-0004-8065-3235.

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